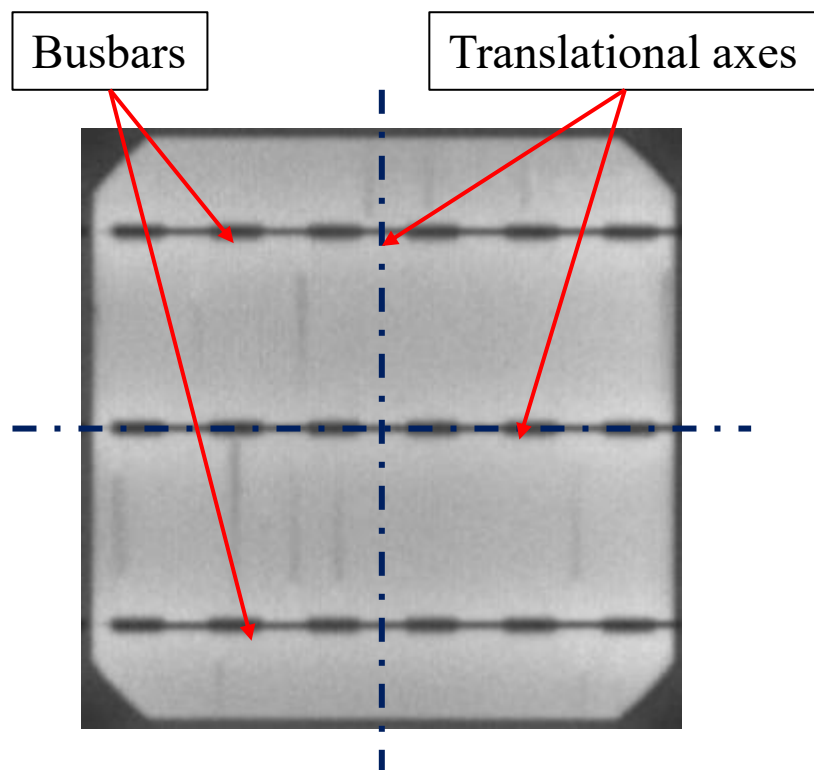
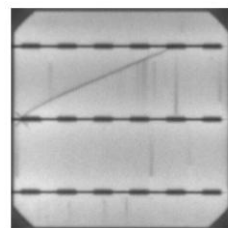
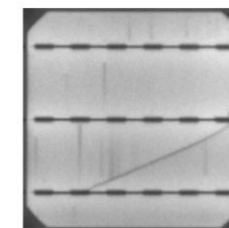
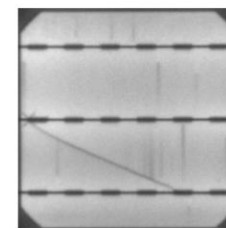
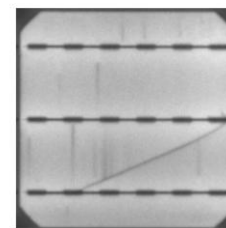
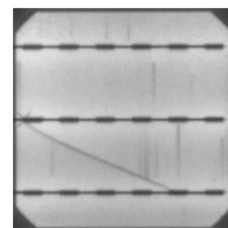




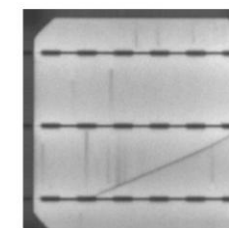
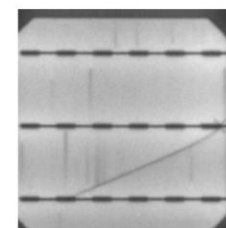
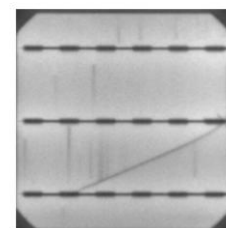
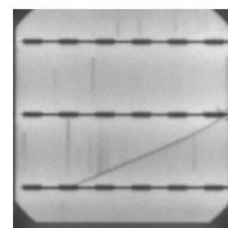
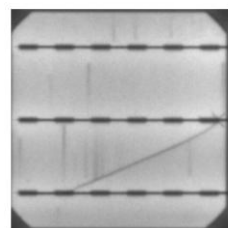
(a) Anatomy of a solar cell



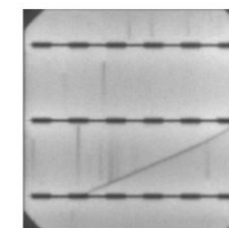
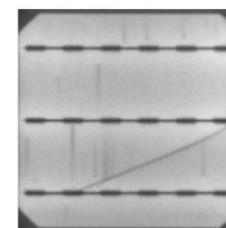
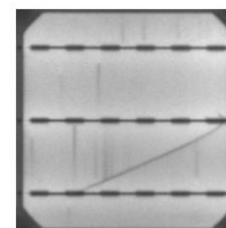
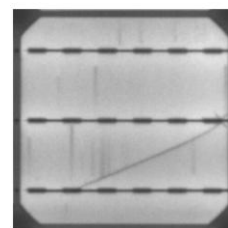
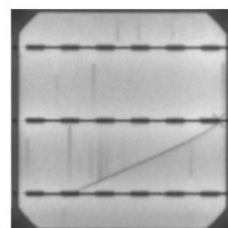
(b) Random horizontal and vertical flip



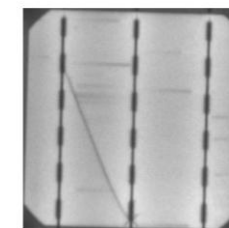
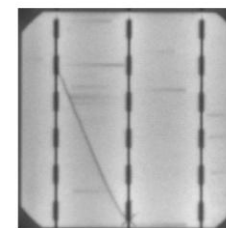
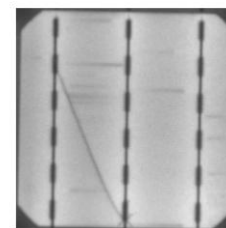
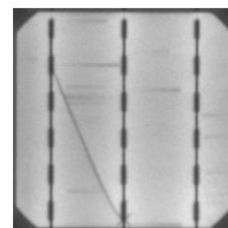
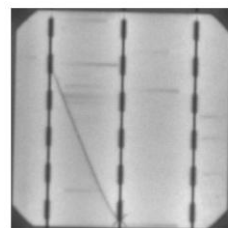
(c) Random horizontal and vertical translation



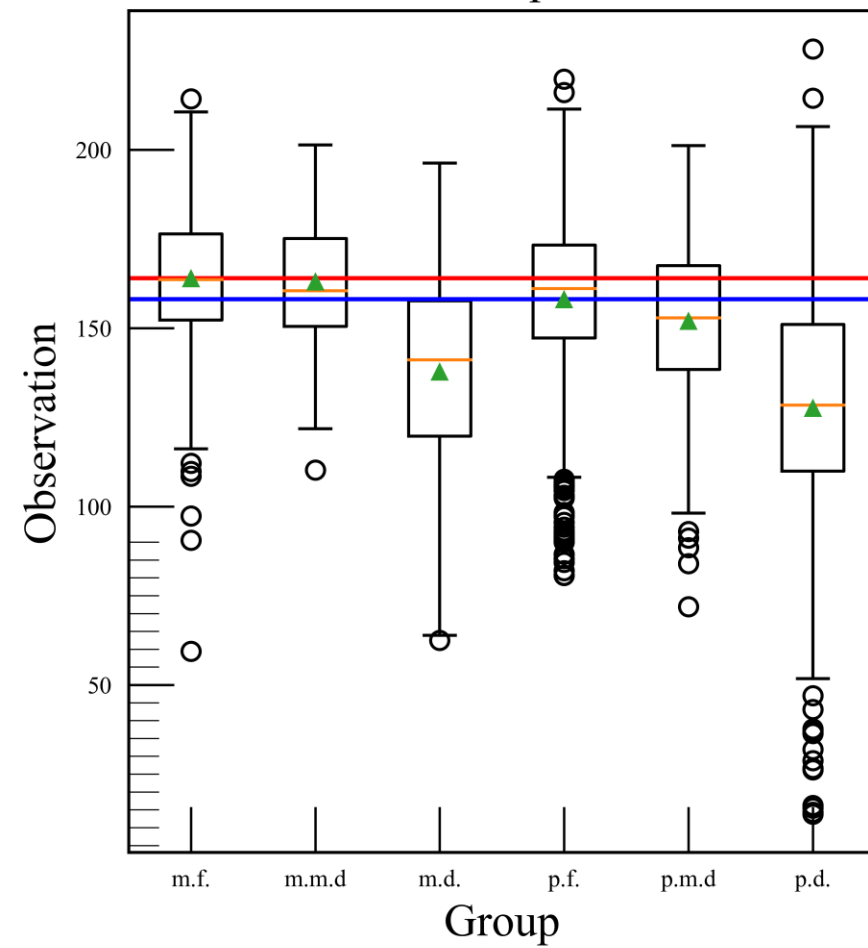
(d) Random zoom



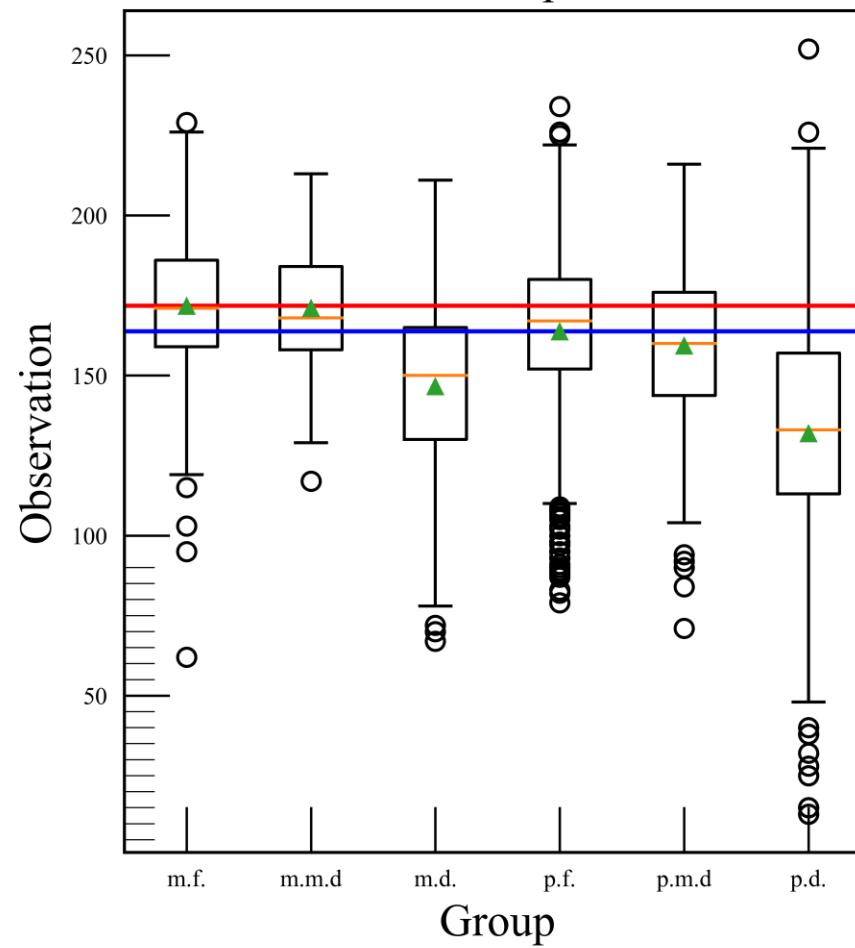
(e) Random rotation by 90°



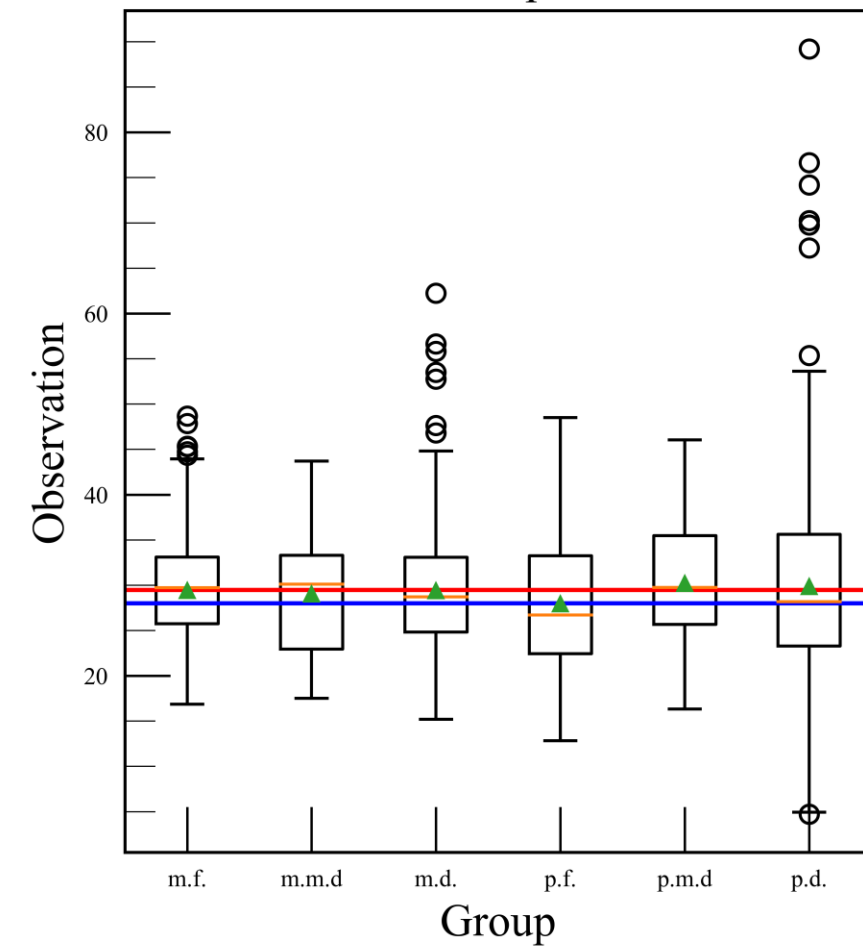
Mean of pixels



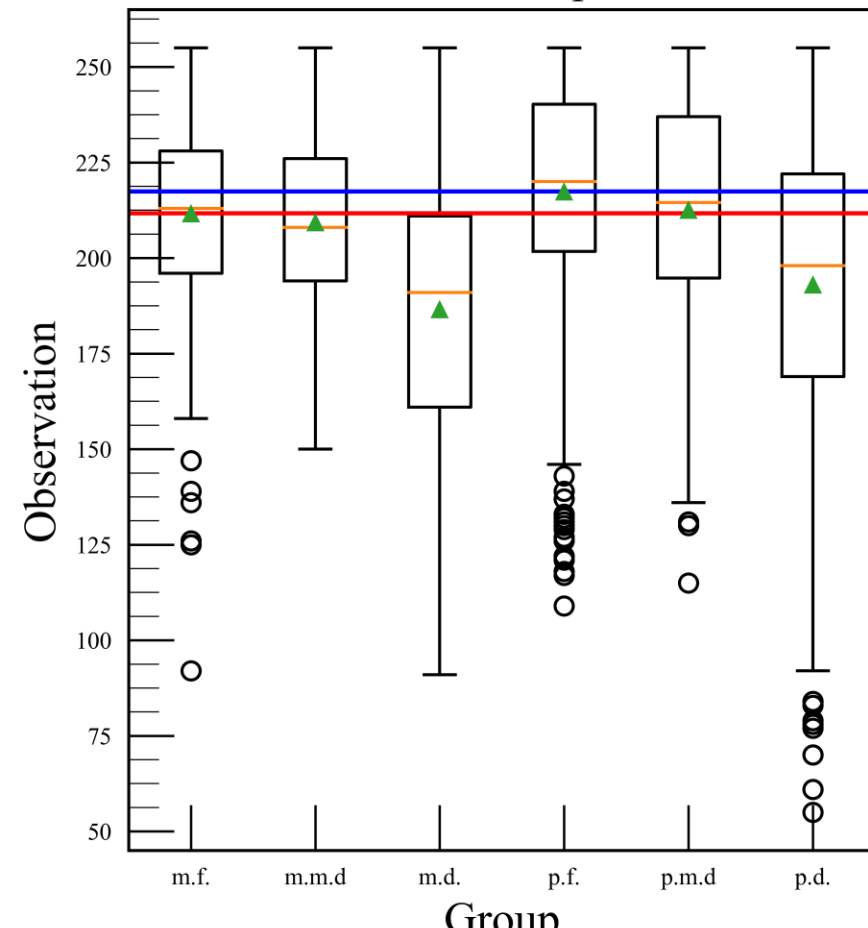
Median of pixels



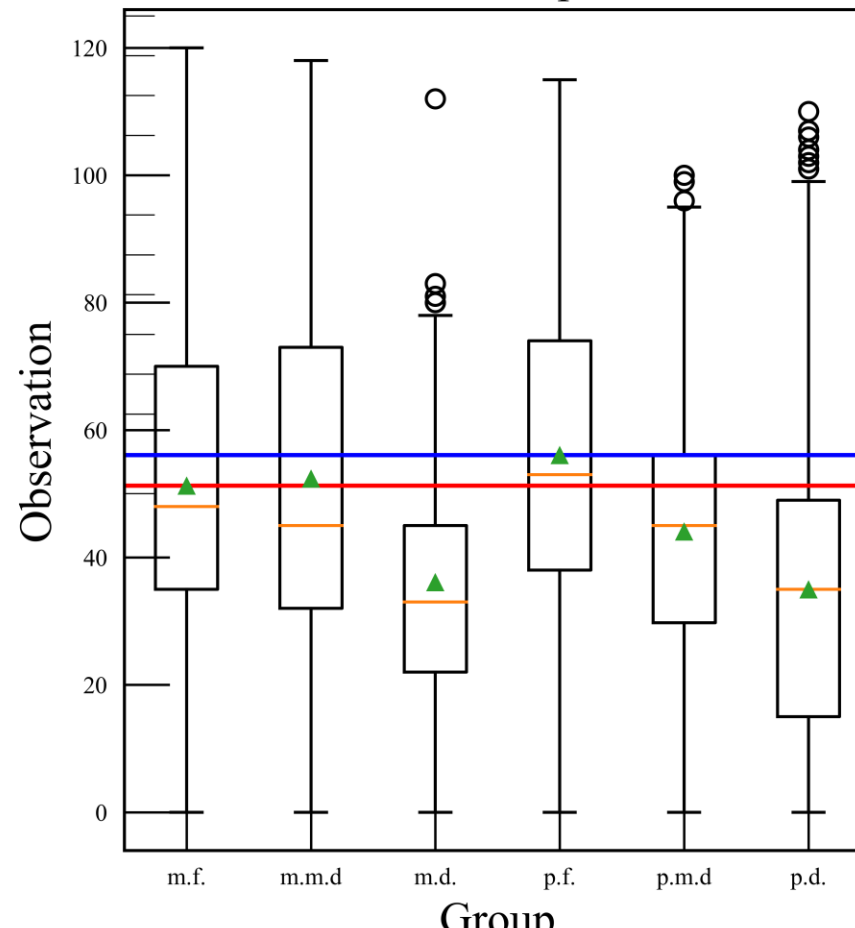
Stdev of pixels



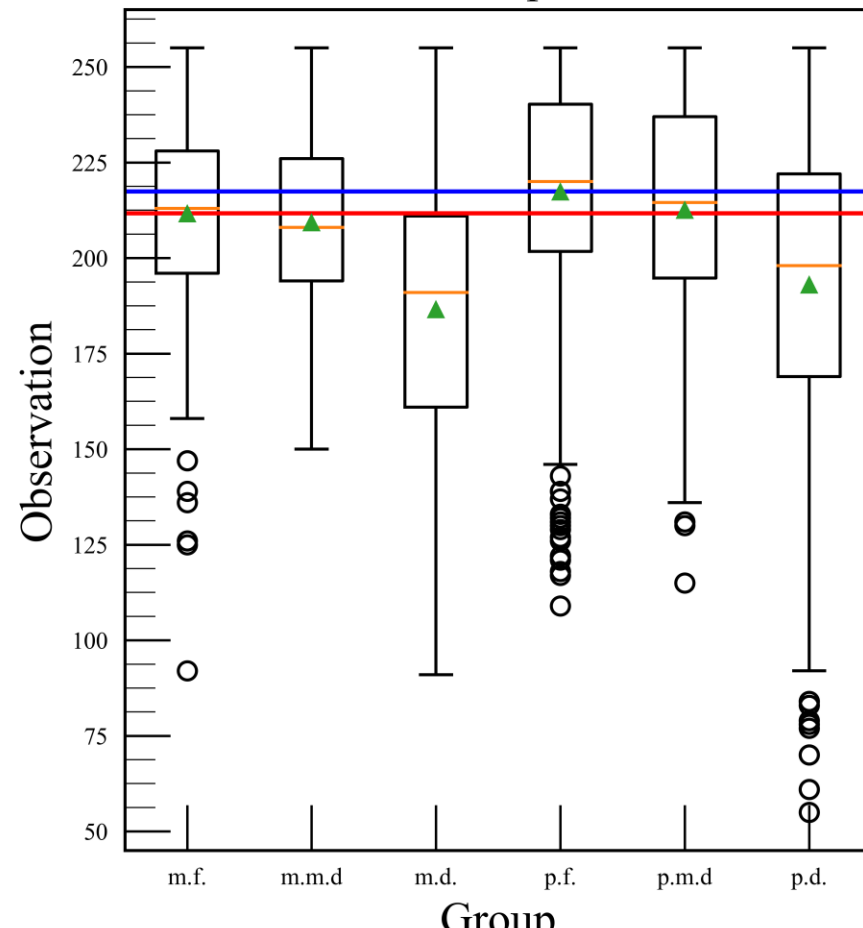
Maximum of pixels



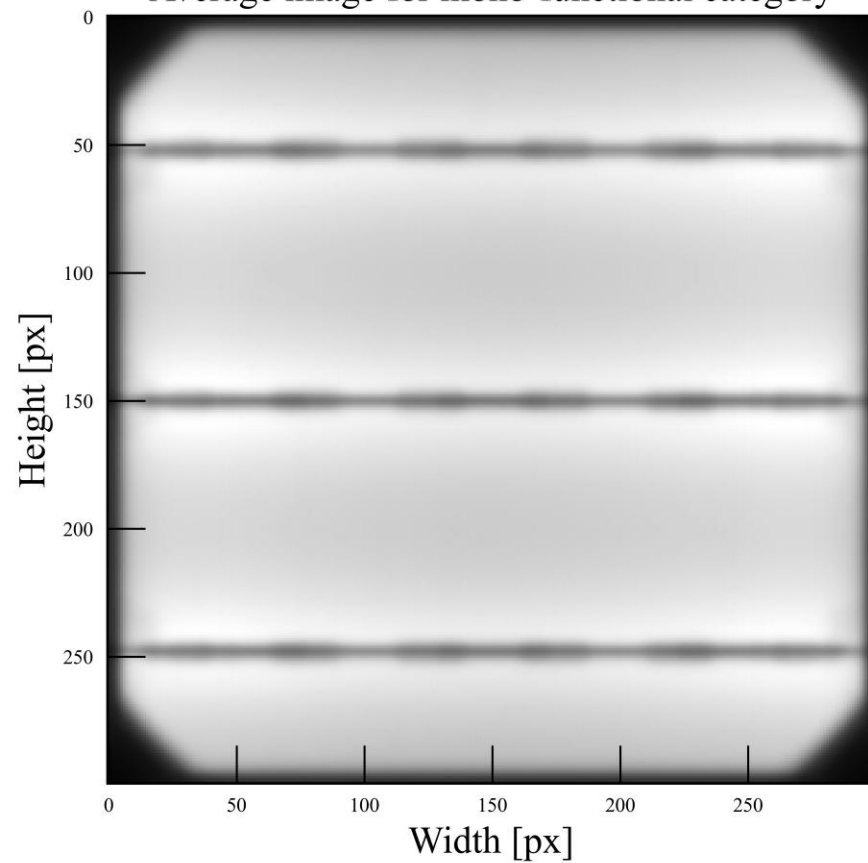
Mininum of pixels



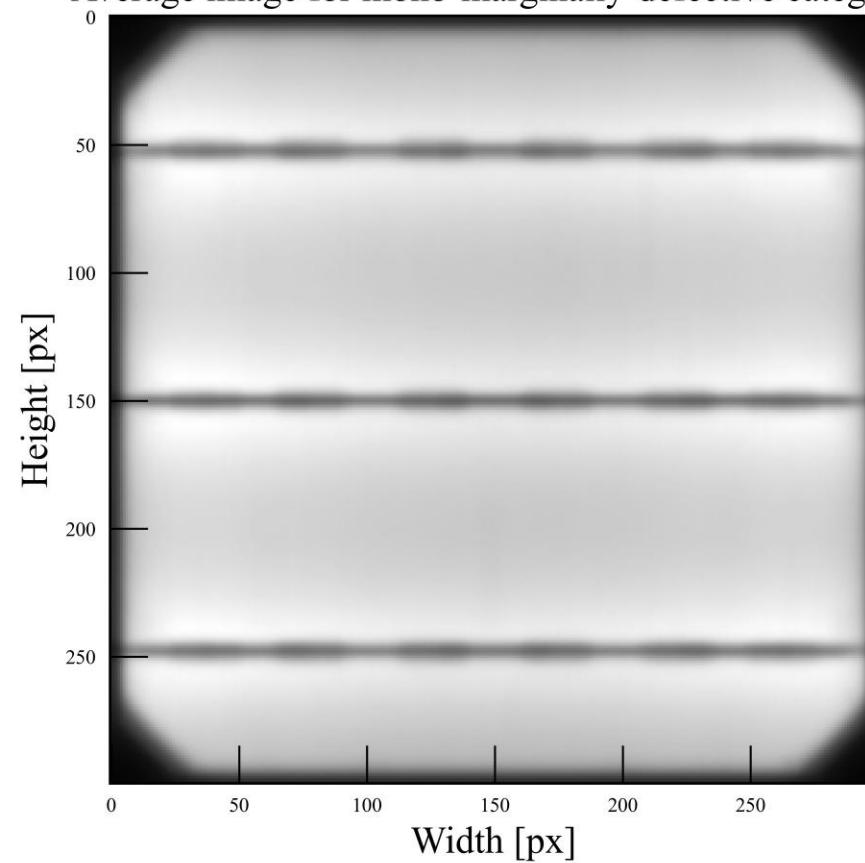
Mode of pixels



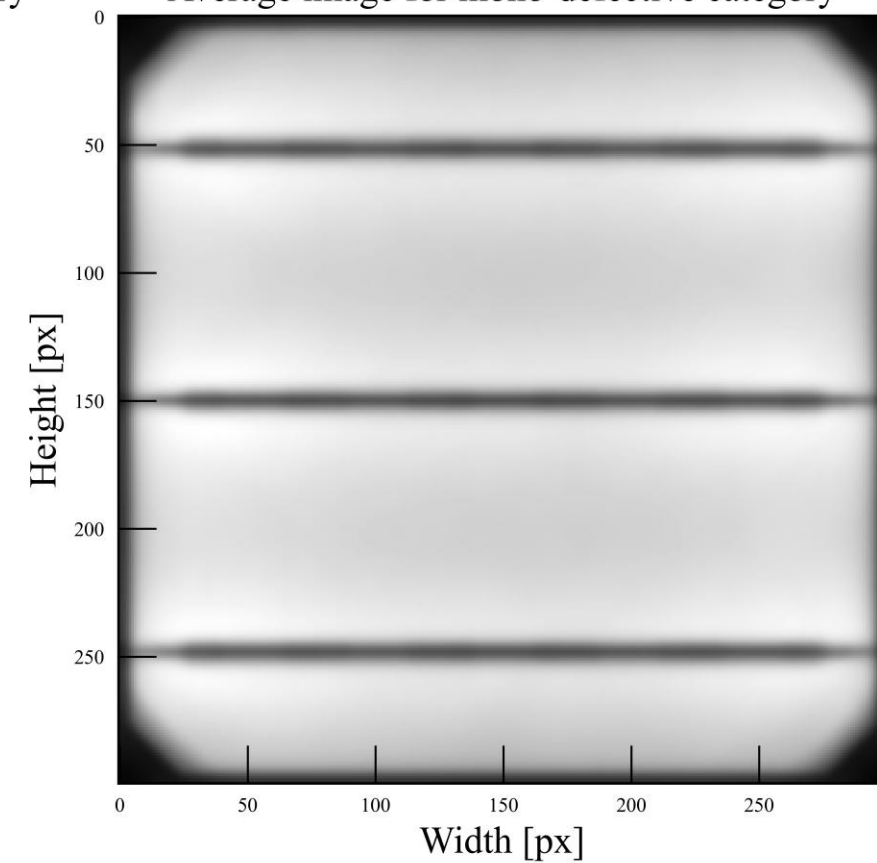
Average image for mono-functional category



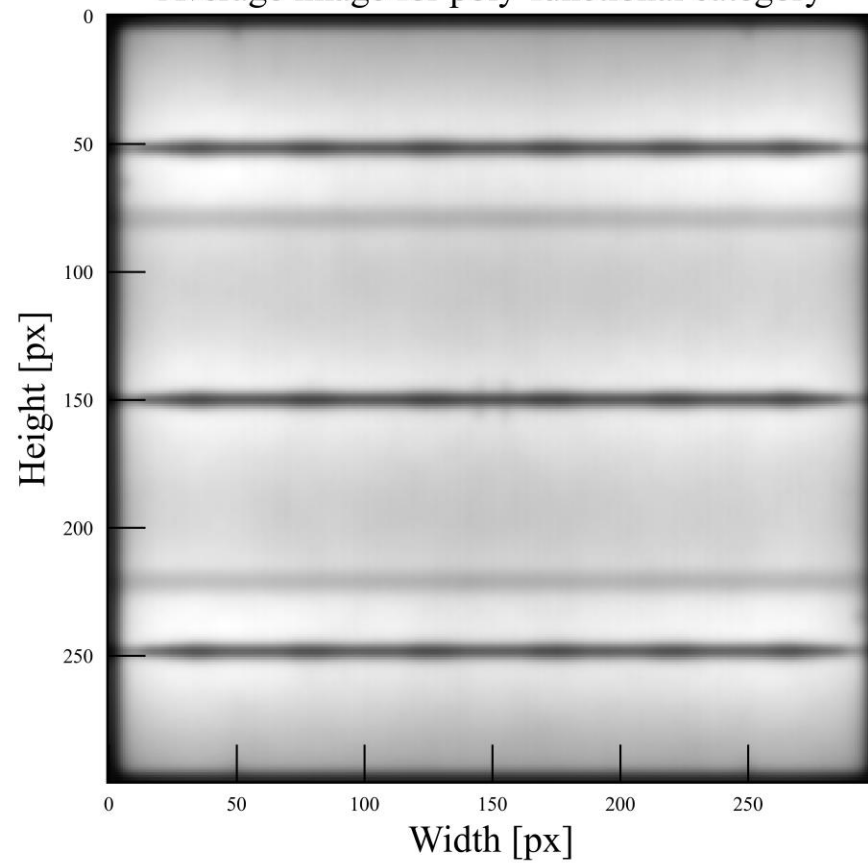
Average image for mono-marginally-defective category



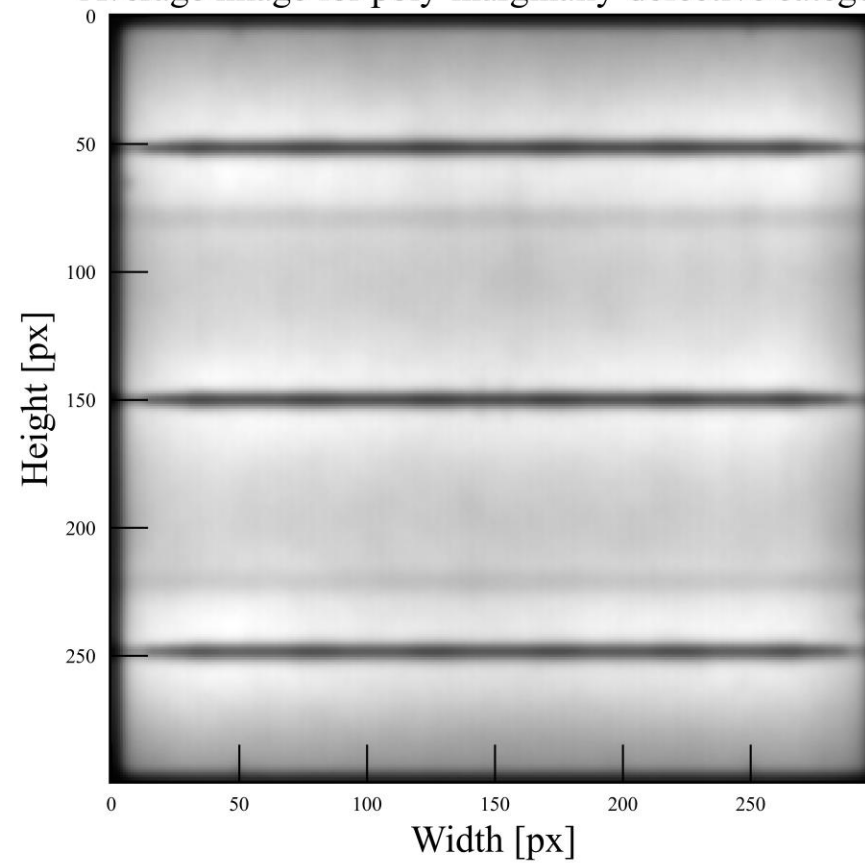
Average image for mono-defective category



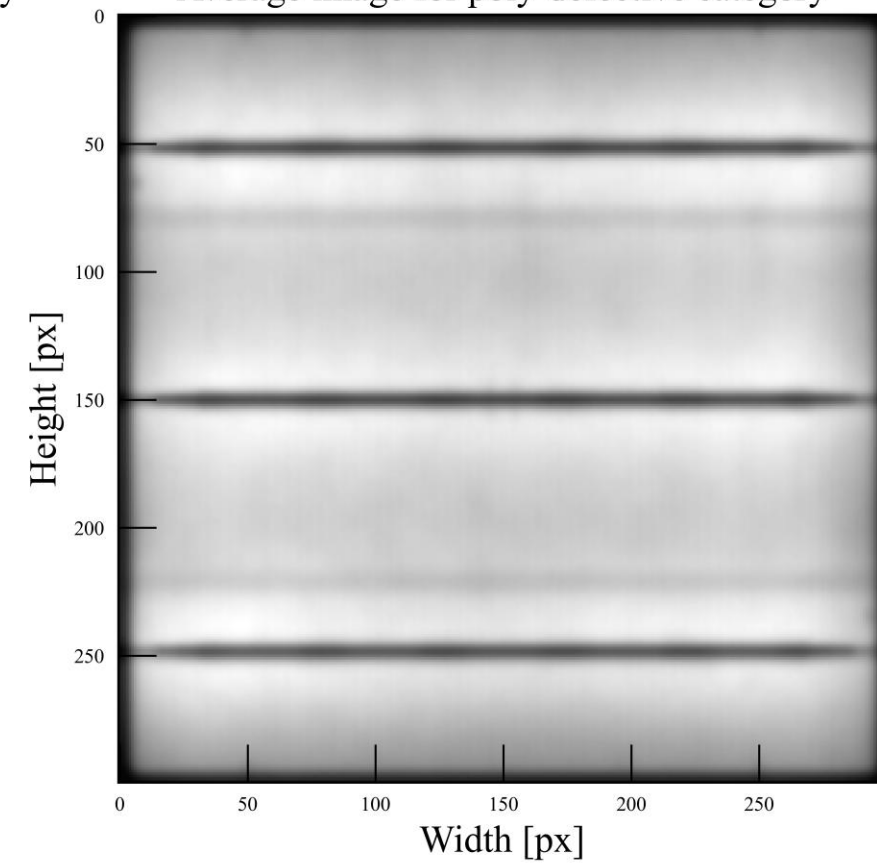
Average image for poly-functional category



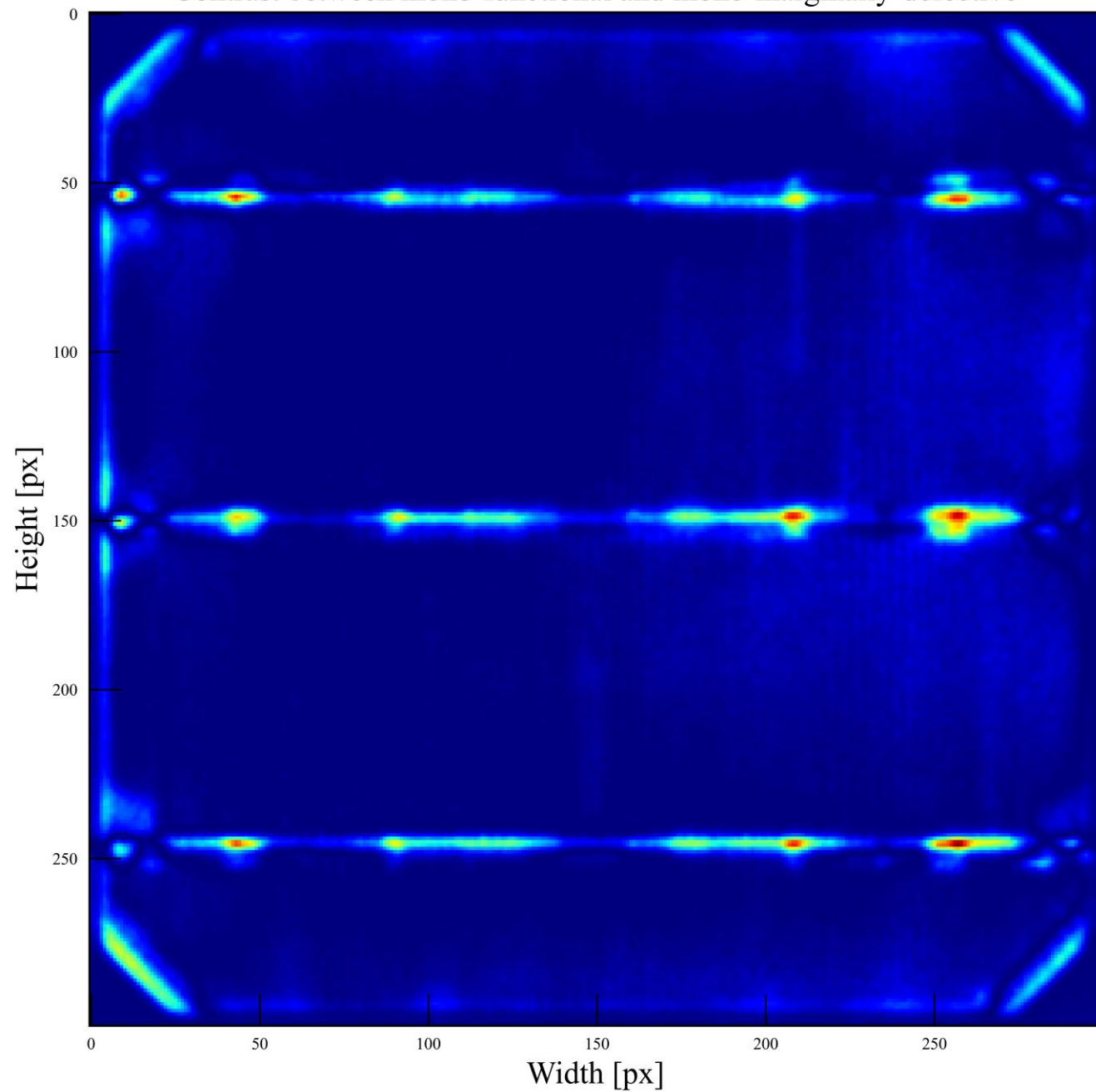
Average image for poly-marginally-defective category



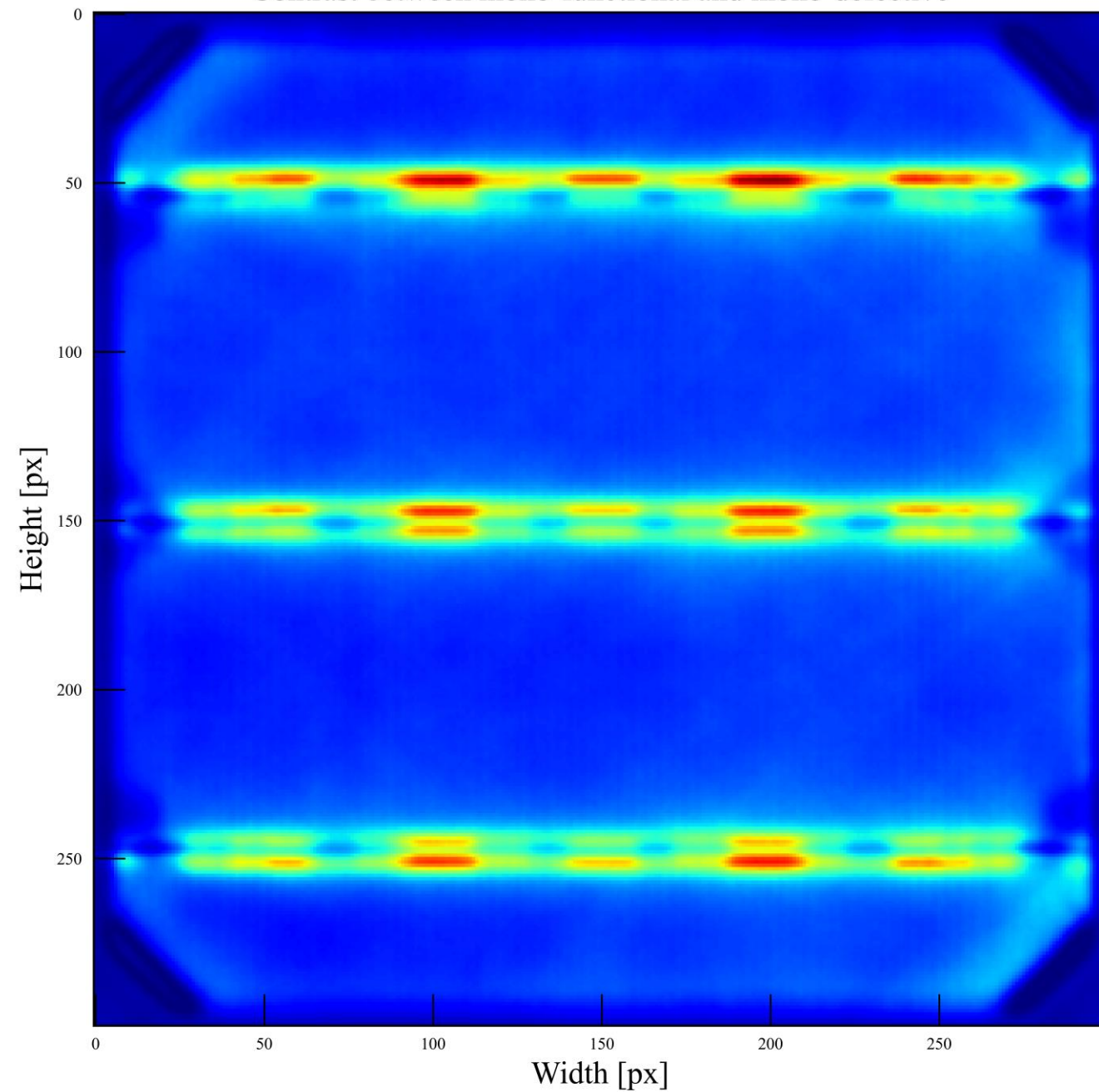
Average image for poly-defective category



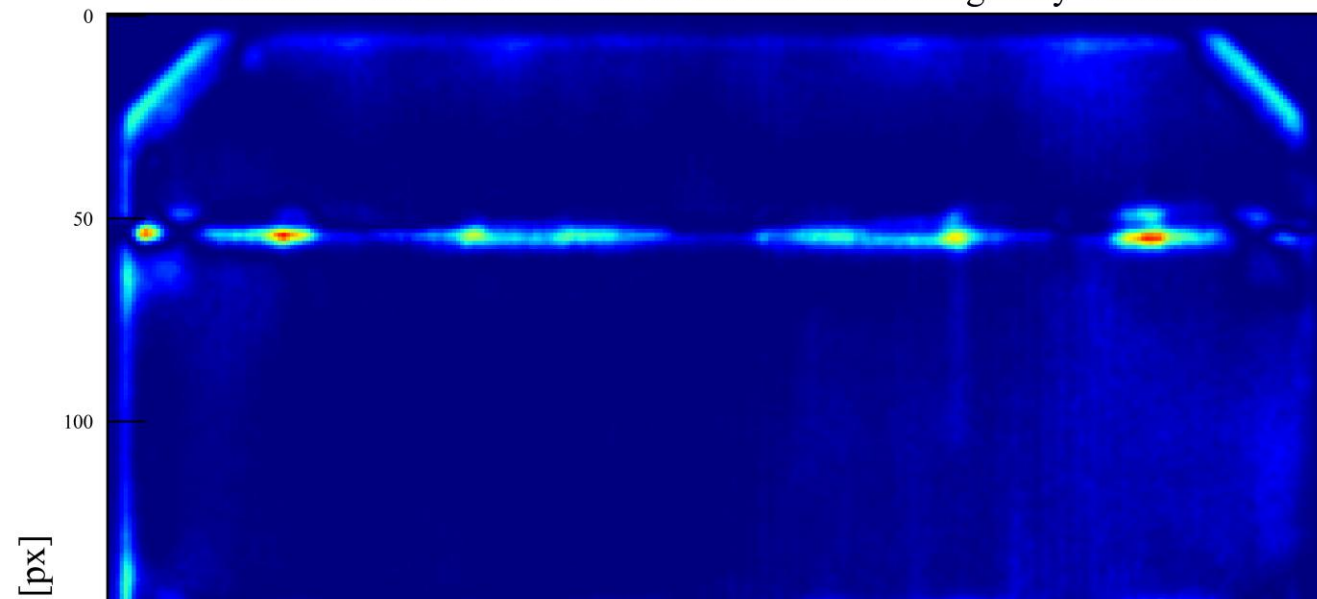
Contrast between mono-functional and mono-marginally-defective



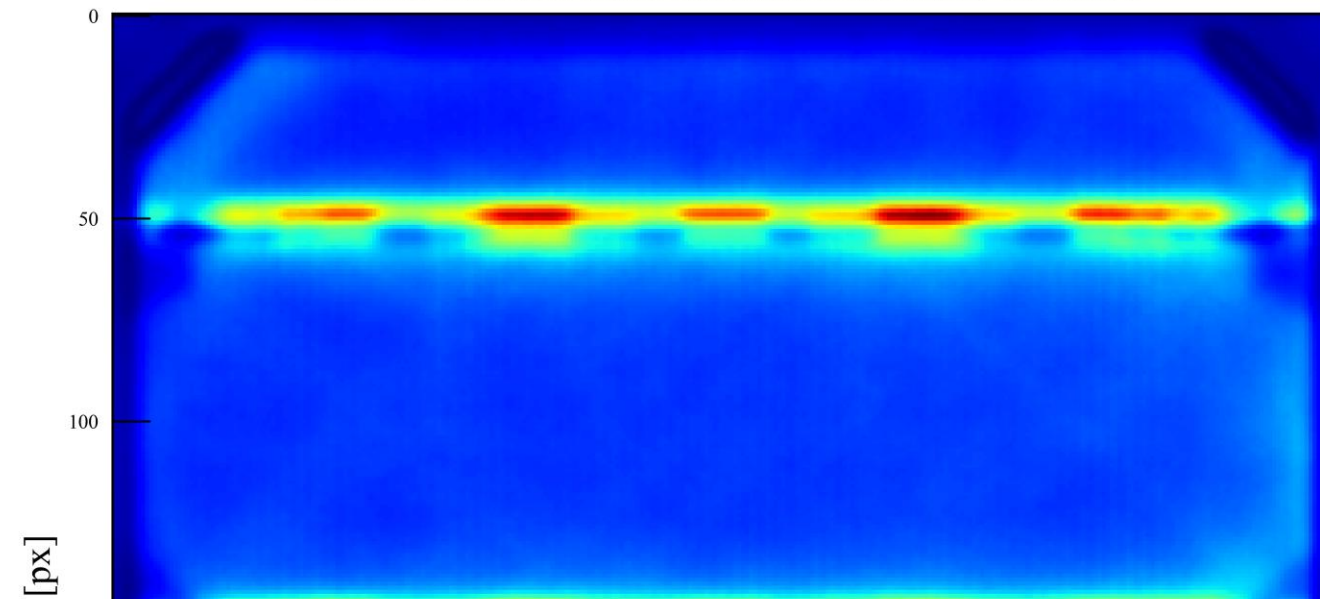
Contrast between mono-functional and mono-defective

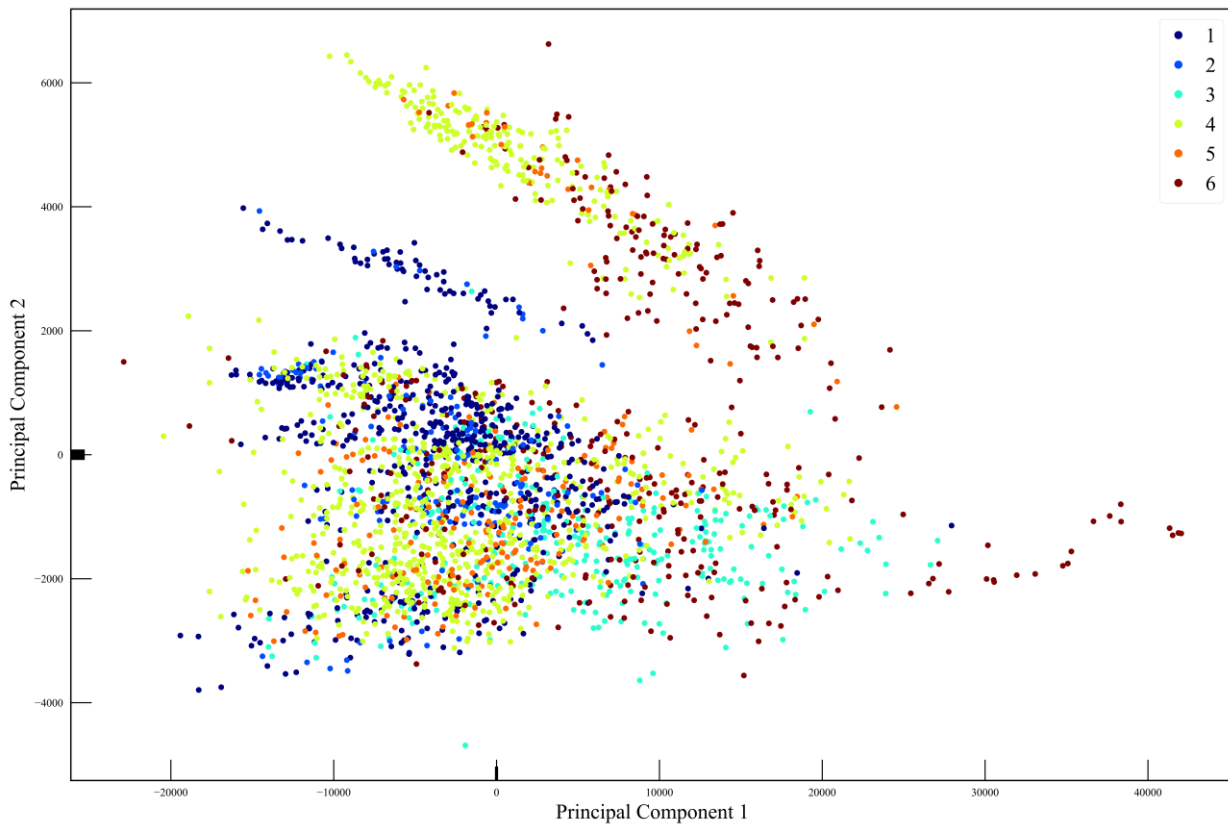


Contrast between mono-functional and mono-marginally-defective

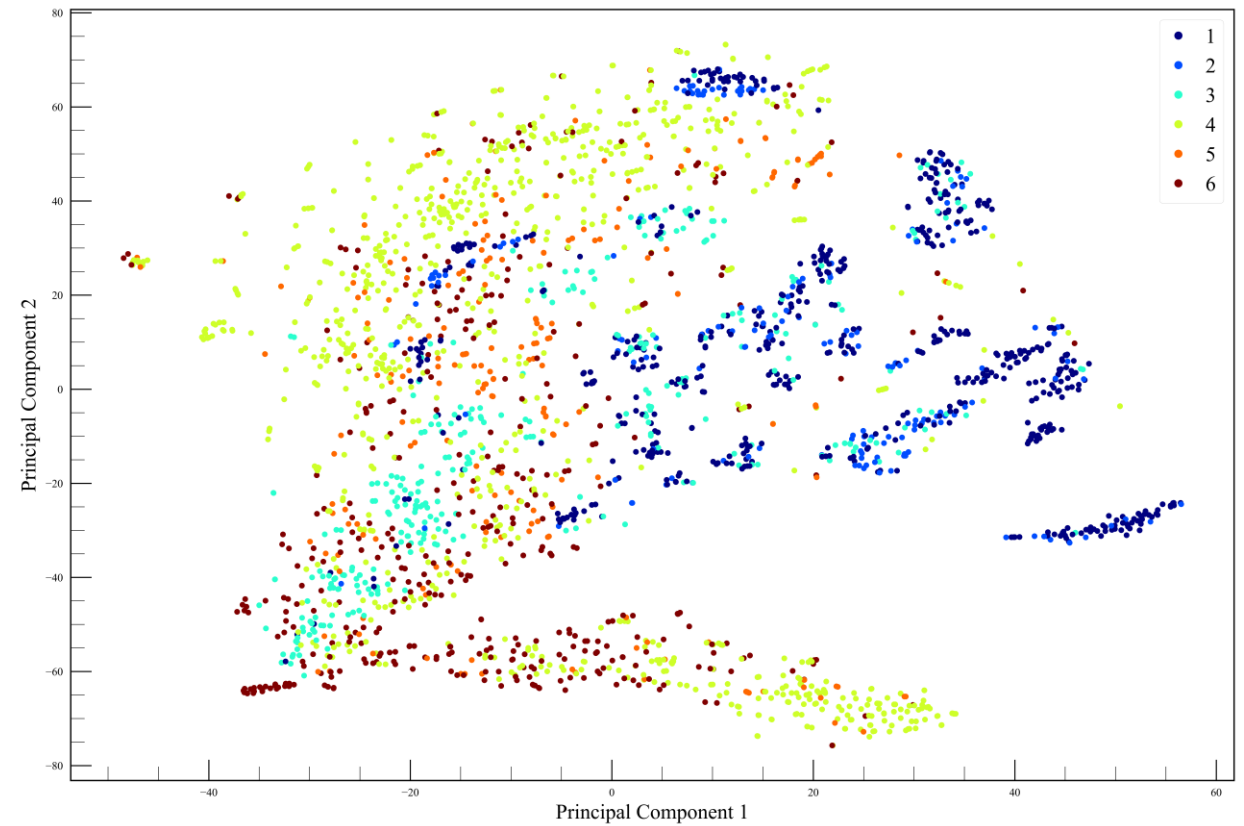


Contrast between mono-functional and mono-defective



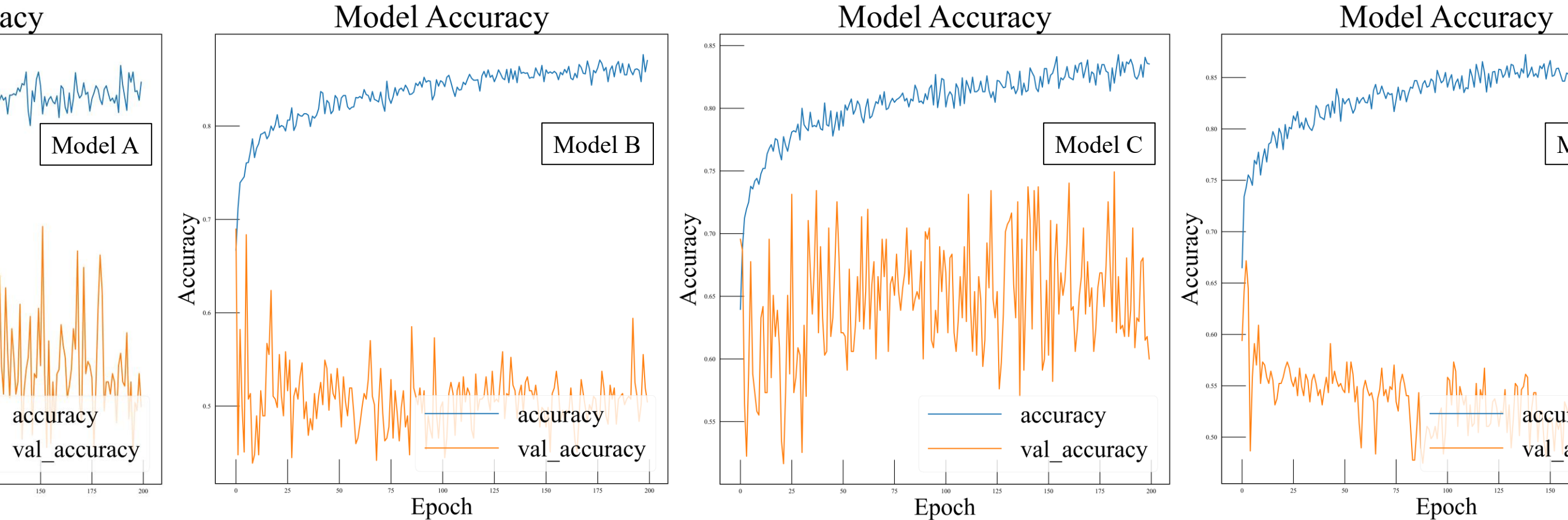


(a) Embeddings using PCA

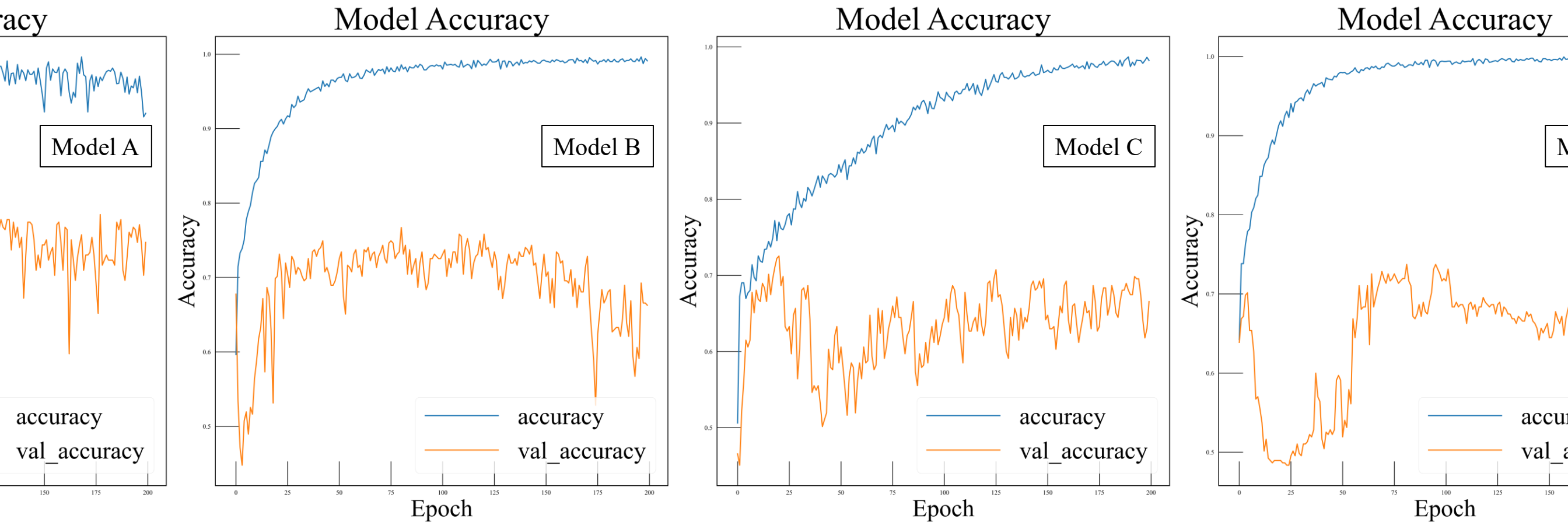


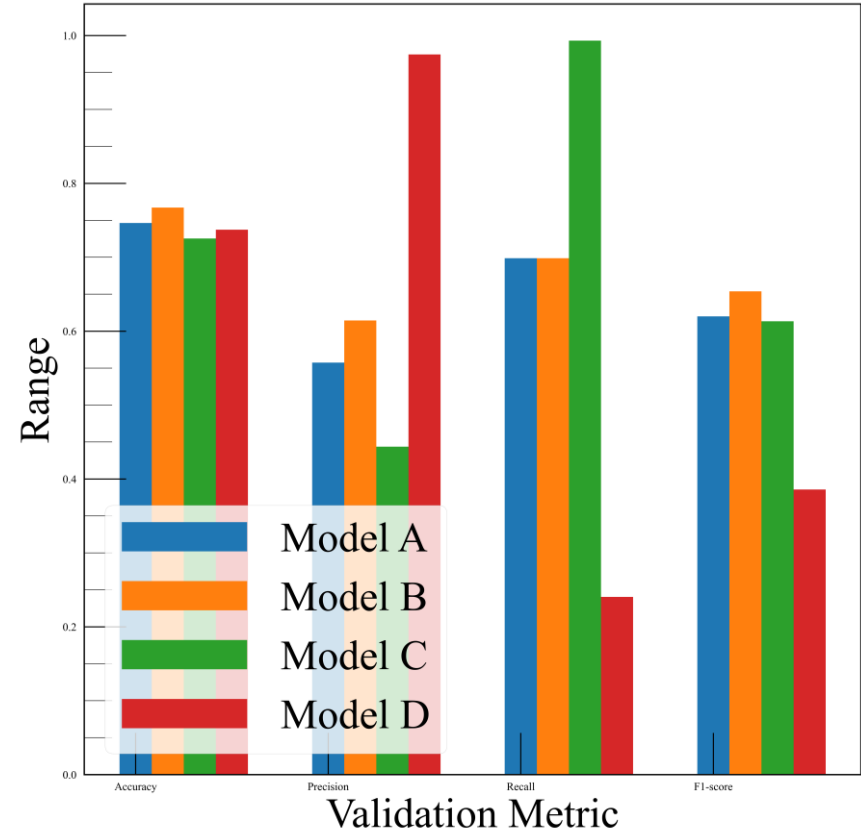
(b) Embeddings using t-SNE

(a) First Phase

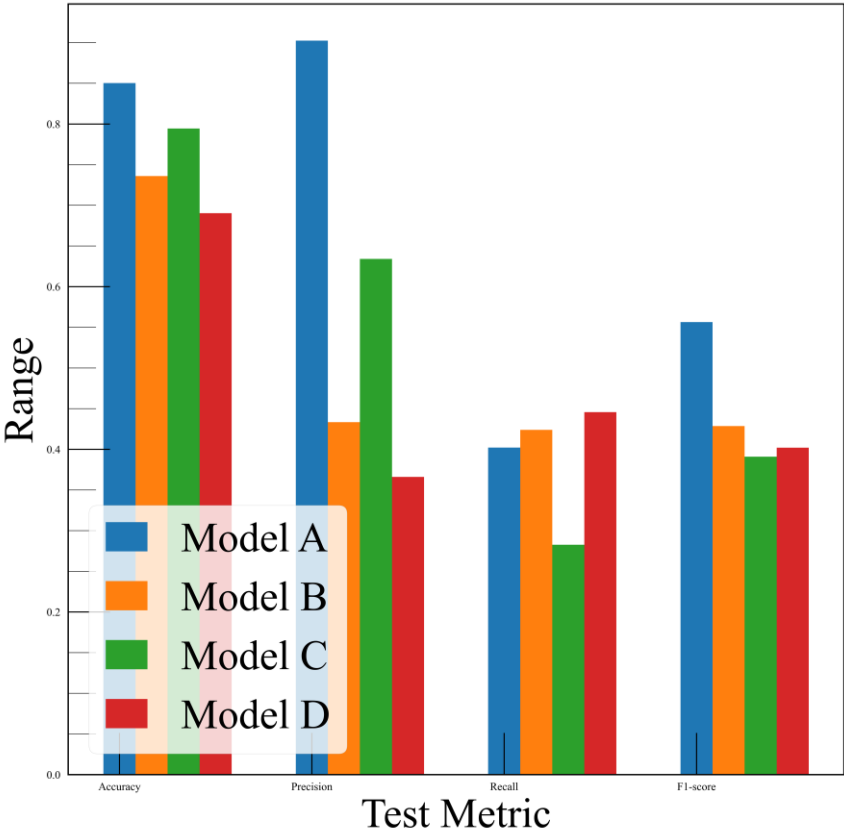


(b) Second Phase

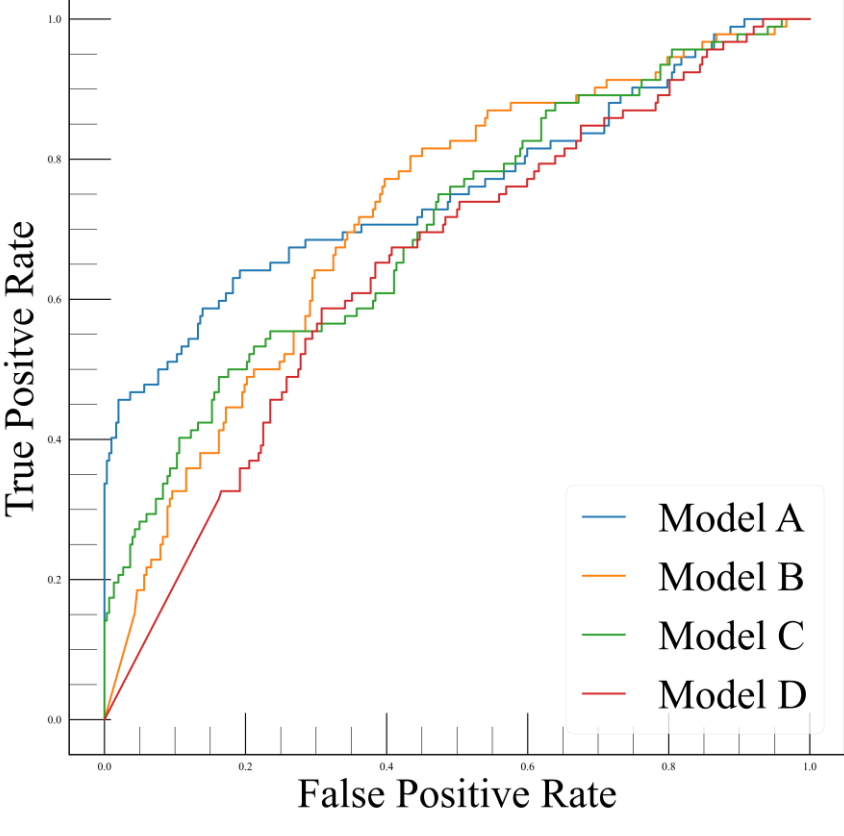




Validation Metric



Test Metric

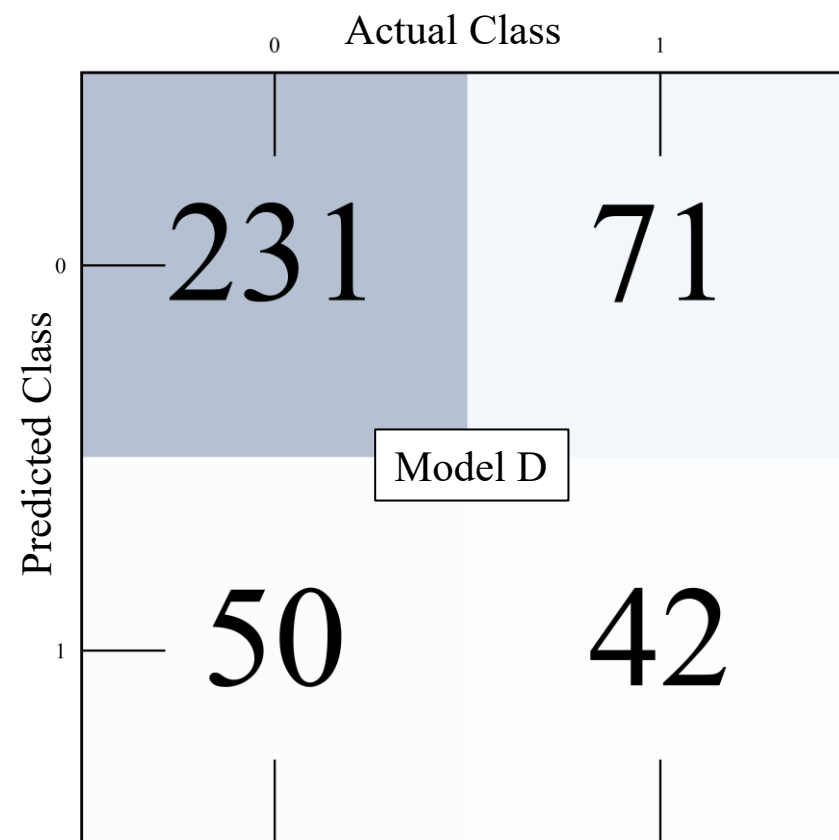
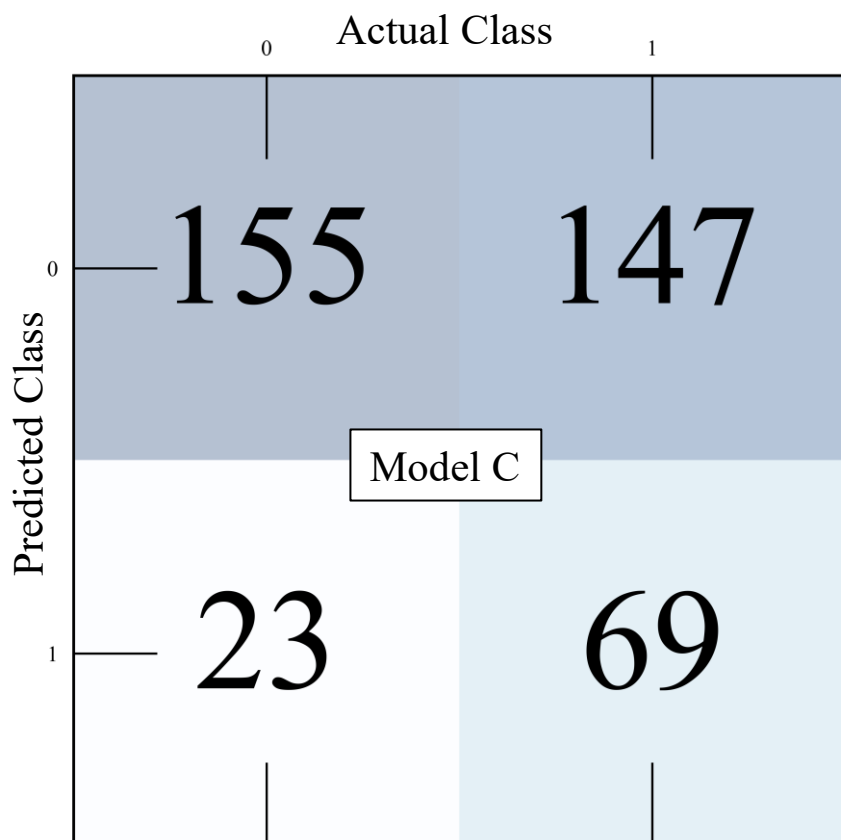
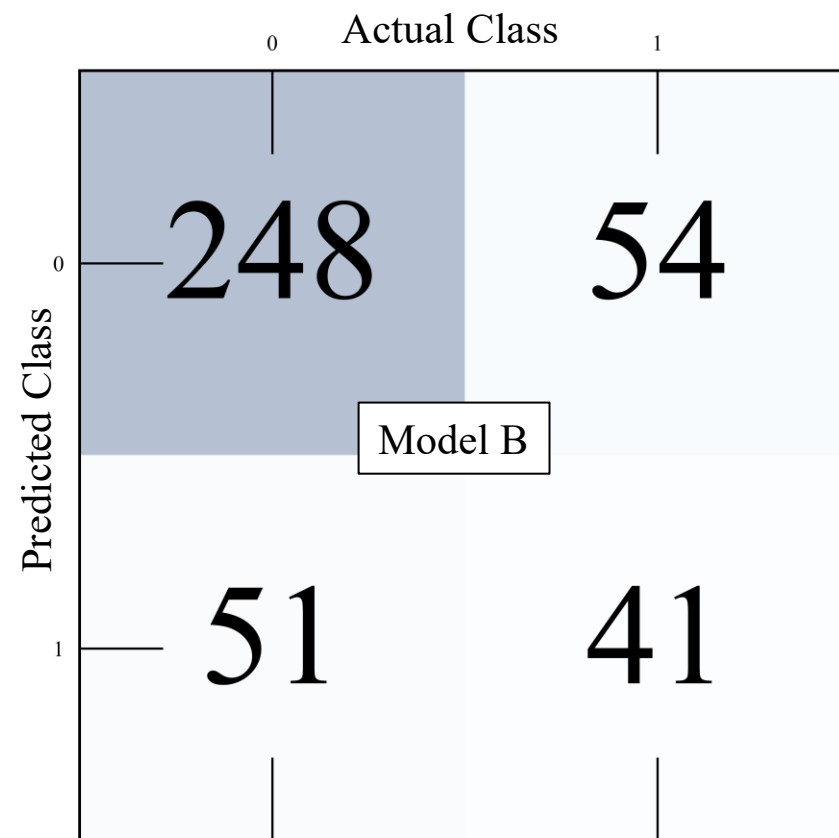
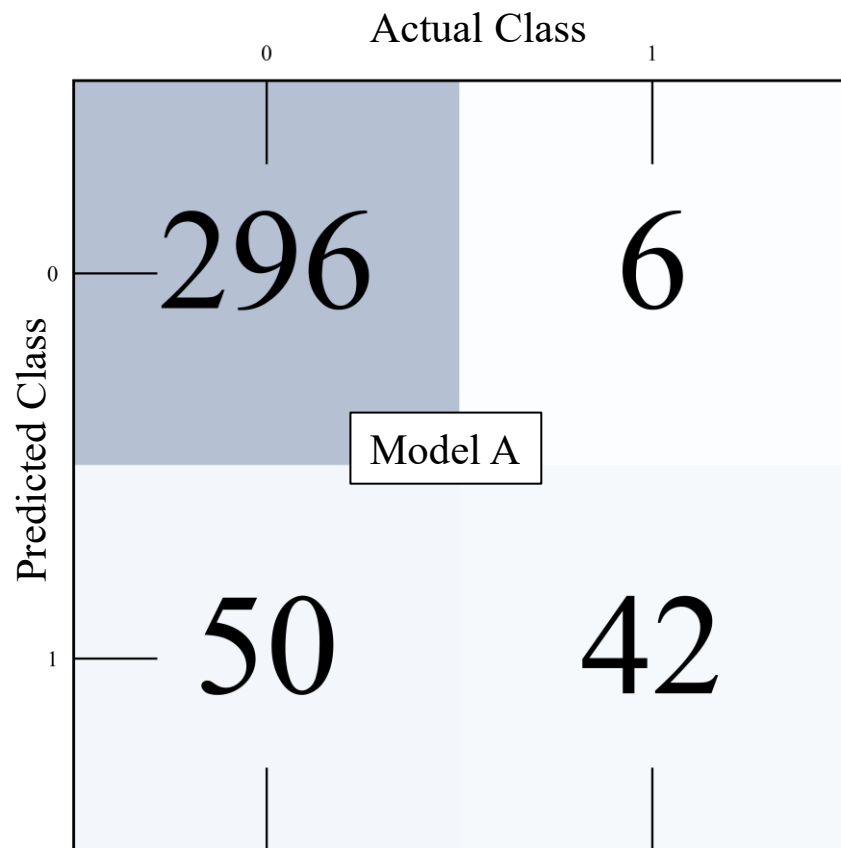


False Positive Rate

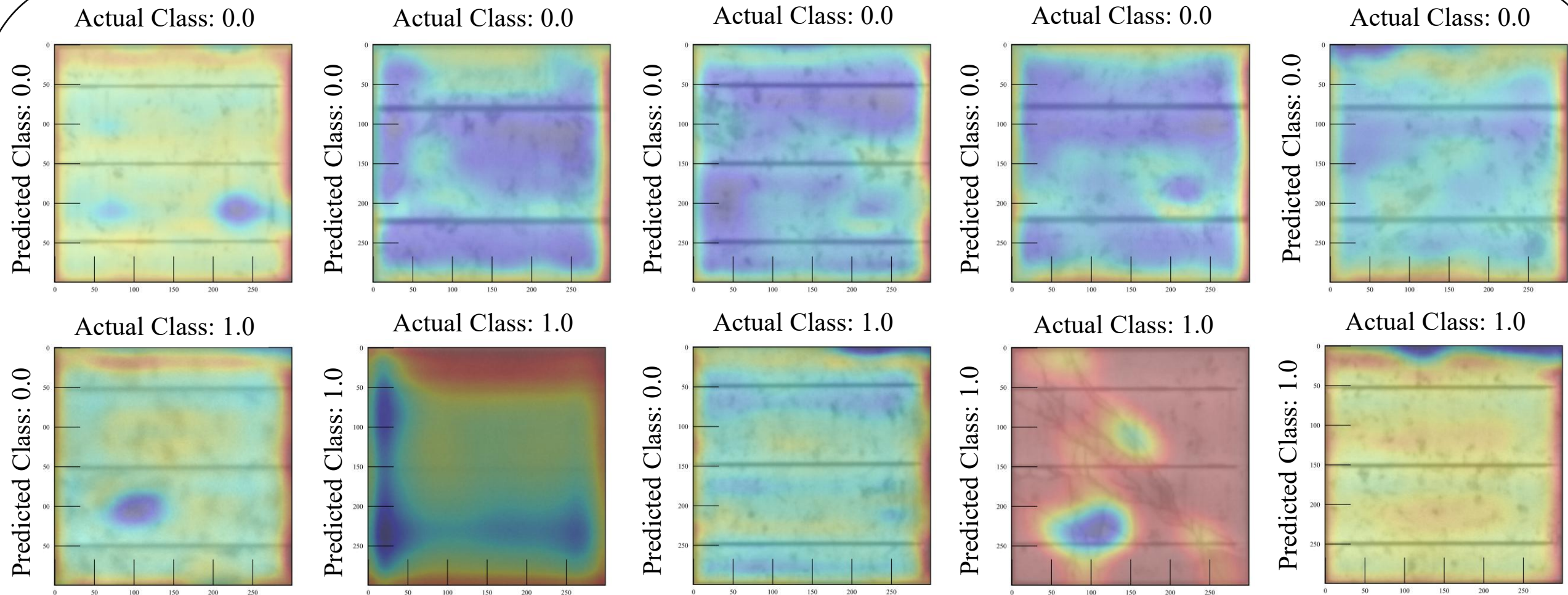
	Model A	Model B	Model C	Model D
Accuracy	0.7463	0.7671	0.7254	0.7373
Precision	0.5574	0.6145	0.4437	0.9745
Recall	0.6986	0.6986	0.9930	0.2405
F1-score	0.6201	0.6538	0.6133	0.3858

	Model A	Model B	Model C	Model D
Accuracy	0.8502	0.7360	0.7944	0.6904
Precision	0.9024	0.4333	0.6341	0.3661
Recall	0.4022	0.4239	0.2826	0.4456
F1-score	0.5564	0.4286	0.3910	0.4020

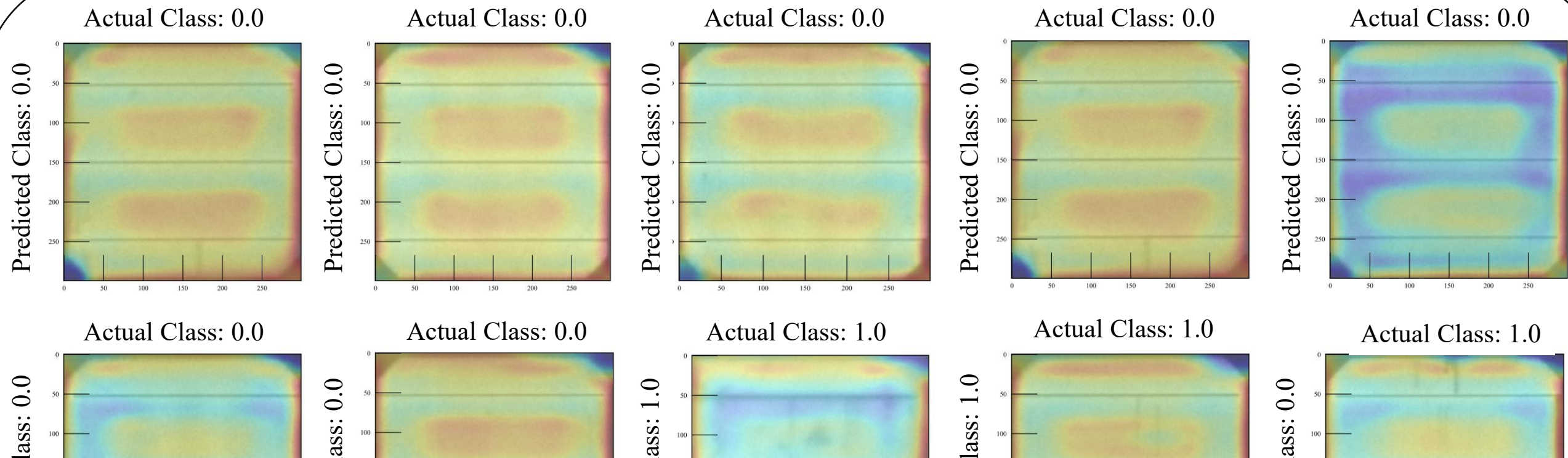
	Model A	Model B	Model C	Model D
AUC	0.7557	0.7186	0.7040	0.6491

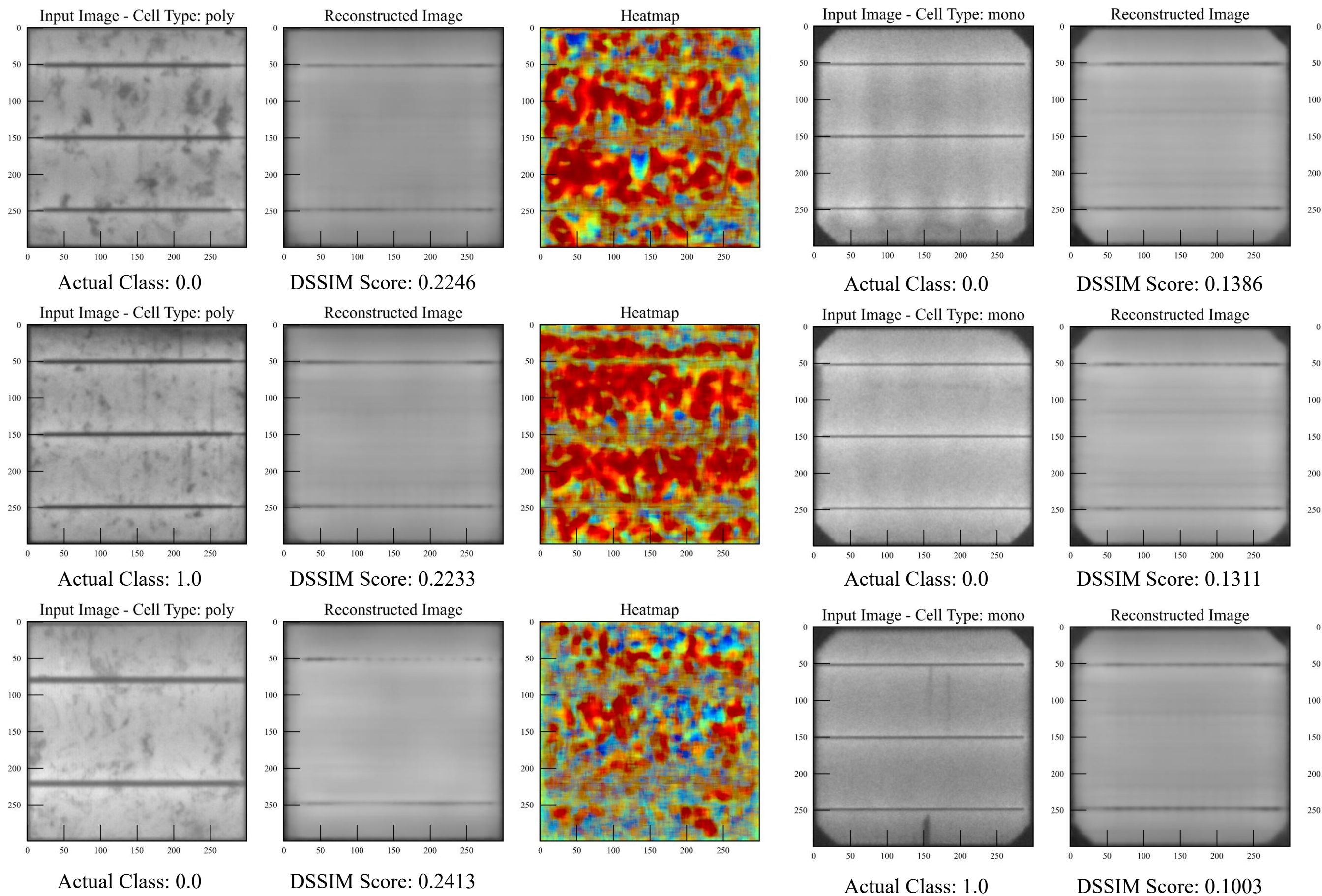


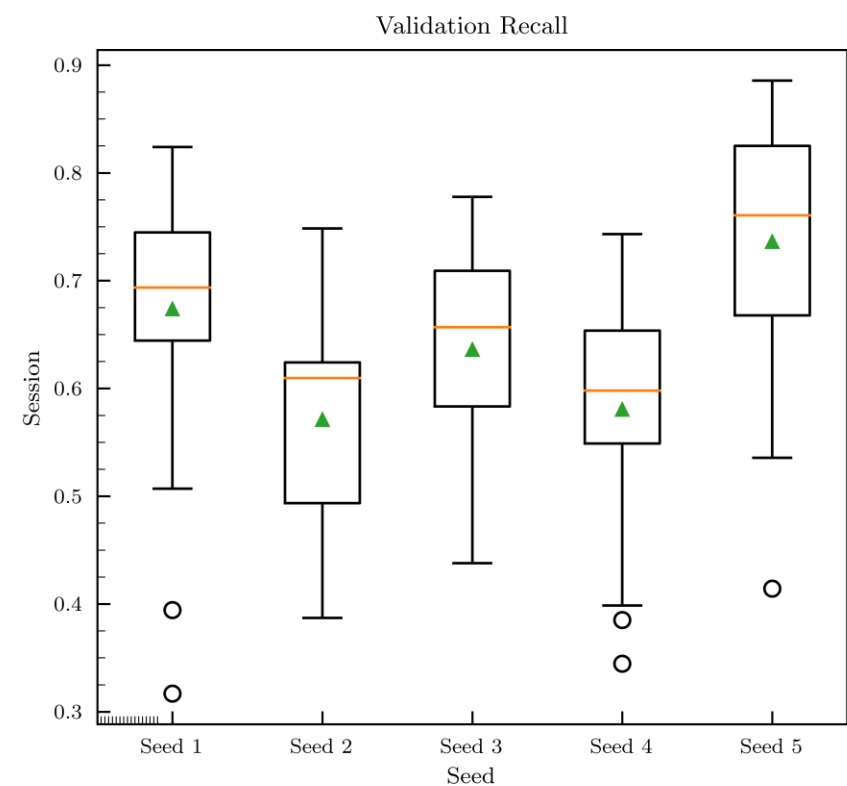
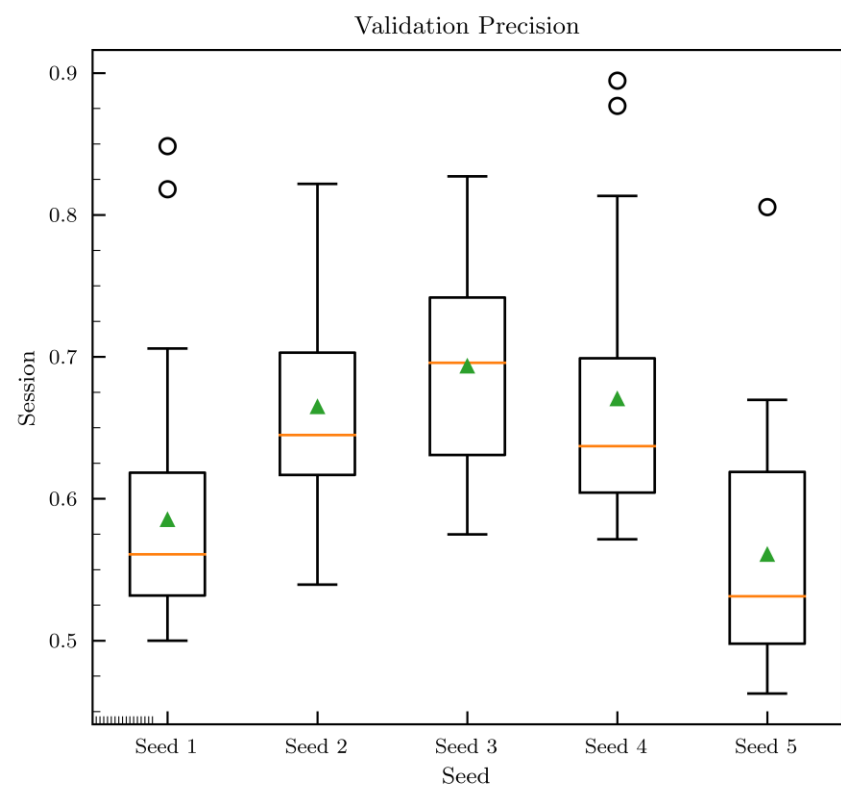
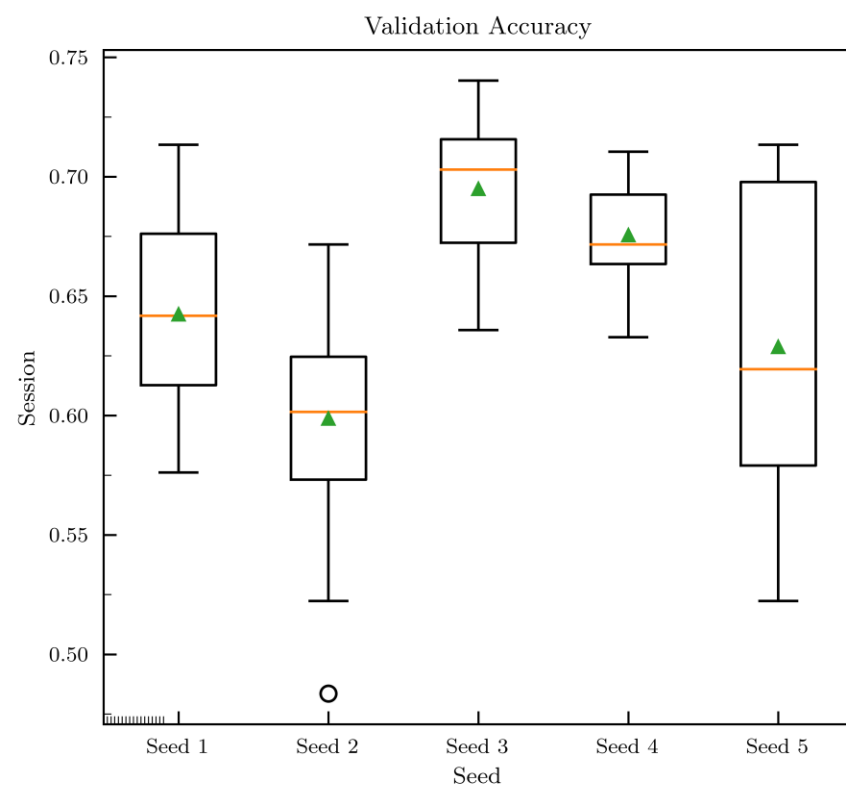
Polycrystalline



Monocrystalline







Validation Metric	Seed 1	Seed 2	Seed 3	Seed 4	Seed 5
Mean accuracy	0.6427	0.5990	0.6952	0.6758	0.6291
Mean precision	0.5855	0.6652	0.6938	0.6707	0.5610
Mean recall	0.6742	0.5716	0.6366	0.5810	0.7367

Brown-Forsythe Test	$H_0: \sigma_1 = \sigma_2 = \sigma_3 = \sigma_4 = \sigma_5$ $H_a: \sigma_1 \neq \sigma_2 \neq \sigma_3 \neq \sigma_4 \neq \sigma_5$	
Accuracy	F = 8.2583	p = 4e − 6
Precision	F = 0.1304	p = 0.9712
Recall	F = 0.2067	p = 0.9344

One-way ANOVA	$H_0: \mu_1 = \mu_2 = \mu_3 = \mu_4 = \mu_5$ $H_a: \mu_1 \neq \mu_2 \neq \mu_3 \neq \mu_4 \neq \mu_5$	
Accuracy*	F = 30.7148	p = 3e − 06
Precision	F = 15.4489	p = 1e − 15
Recall	F = 13.3740	p = 2 − 09